View Meta-Reviews

Paper ID
850

Paper Title
H2S-NN: Heterogeneous Heterophilic Sheaf Neural Network

Track Name
Full Paper (Main Track)

META-REVIEWER #1

META-REVIEW QUESTIONS

1. Meta-review
This paper proposes H2S-NN, a sheaf neural network for heterogeneous heterophilic graphs. It keeps concrete meta-path instances as path-supported 2-cells rather than collapsing them into endpoint-level relations, and adds AMPSS for meta-path selection and SheafCovPool for graph-level readout. The reviewers find the 2-cell idea novel and the construction principled, and appreciate the evaluation on node classification, graph classification and link prediction.

However, the reviewers are split, both high-confidence reviewers recommend weak reject, and even the positive reviewers note that key claims need better support. First, the heterophilic setting is not established. Only IMDB is strongly heterophilic among the node-classification datasets, the graph-classification benchmarks are standard TU datasets whose labels are reinterpreted as types, and no analysis relates the gains to the degree of heterophily. Second, the contribution of the 2-cells is not isolated. The face term reaches node representations only through the closure edges, and the ablations do not separate it from schema selection, instance weighting and added capacity. Third, the paper does not analyze how the cost grows with graph size and the number of retained path instances. Reviewers also question the evaluation protocol, e.g., baseline tuning and HetSheaf results quoted from the original paper.

I therefore recommend rejection and encourage the authors to address the reviewers' detailed comments in a future submission.